Pediatric ED Care Pathways: A Comprehensive Inventory From America's Leading Children's Hospitals

The most actionable, publicly accessible pediatric emergency clinical pathways come from four institutions — CHOP, Seattle Children's, Texas Children's, and Nationwide Children's — which together publish over 250 free, evidence-based protocols covering the full spectrum of pediatric emergency conditions. These pathways, combined with PECARN clinical decision rules and AAP clinical practice guidelines, form a robust evidence base that community and rural EDs can directly adapt. The landscape has matured dramatically since 2020, with landmark publications including the AAP 2021 febrile infant guideline, the 2024 prospective validation of PECARN head injury and abdominal trauma rules, and multi-center quality improvement collaboratives demonstrating that pathway implementation reduces unnecessary testing, shortens hospital stays, and improves outcomes — even in non-academic settings. This report catalogs the best available pathways across four core conditions (asthma, bronchiolitis, fever, and trauma), maps the national infrastructure supporting pathway dissemination, and identifies the most practical resources for community hospitals lacking subspecialty pediatric emergency access.

The pathway library landscape and where to find free protocols

Four major children's hospitals maintain publicly accessible clinical pathway libraries that serve as the backbone for community hospital adaptation nationwide.

CHOP's Clinical Pathways Program is the single most comprehensive publicly available resource, hosting **90 ED-specific pathways** plus 87 inpatient, 76 ICU, and 46 primary care pathways — all freely accessible at chop.edu/pathways-library without login. Each pathway includes algorithms, order sets, quality metrics, and provider resources. Children's Hospital of Philadelp… CHOP pathways are developed by multidisciplinary teams Children's Hospital of Philadelp… within the Center for Healthcare Quality & Analytics and are designed to standardize care for approximately 80% of the relevant patient population while preserving clinician judgment for atypical presentations. Texas Children's Children's Hospital of Philadelp…

Seattle Children's Clinical Standard Work pathways (seattlechildrens.org/healthcare-professionals/community-providers/pathways/) encompass over 70 conditions as downloadable PDFs, built using formal GRADE evidence review methodology. These pathways receive approximately **45,000 downloads annually**, with roughly half from outside Seattle Children's, including from 15 foreign countries. Children's Hospital Association A 2016 interrupted time series analysis across 15 Seattle pathways demonstrated a significant halt in rising costs (–$155/month slope difference) and decreased length of stay (–0.03 days/month) without adverse effects on patient functioning or readmissions. PubMed

Texas Children's Evidence-Based Outcomes Center (texaschildrens.org) publishes over 90 clinical standards as free PDFs, using GRADE methodology with a health disparities framework integrated into each pathway. **Cincinnati Children's James M. Anderson Center** (cincinnatichildrens.org) pioneered evidence-based pediatric pathways beginning in 1996, publishing 29+ formal guidelines through its LEGEND (Let Evidence

Guide Every New Decision) system (ResearchGate) — though some current pathways have migrated to internal EHR tools. **Nationwide Children's Hospital** maintains a well-organized clinical pathway hub with separate ED and inpatient PDFs for multiple conditions. (Nationwide Children's Hospital) (Nationwide Children's Hospital)

A 2024 survey of 111 hospitals in the Pediatric Research in Inpatient Settings (PRIS) network found that **63% of children's hospitals now have a formal clinical pathway program**, (American Academy of Pediatrics) though over half reported minimal or no systematic measurement of pathway outcomes — highlighting a persistent gap between pathway creation and rigorous evaluation.

Acute asthma exacerbation: score-driven protocols converging on dexamethasone and MDI-first strategies

Pediatric asthma exacerbation pathways have achieved remarkable consensus on key treatment principles while differing primarily in which severity scoring tool they employ. Every major pathway now emphasizes **systemic corticosteroids within 60 minutes of arrival**, (Nationwide Children's Hospital) metered-dose inhaler (MDI) preference over nebulizer for mild-to-moderate disease, and structured phase-based treatment escalation.

Severity scoring remains institutionally fragmented. No single tool has achieved universal adoption. CHOP uses the **Pediatric Asthma Severity Score (PASS)**, (Children's Hospital of Philadelp…) a 3-item tool (wheezing, work of breathing, prolongation of expiration) scored 1–3 per item, (MDCalc) validated by Gorelick in 2004. Seattle Children's developed a custom **Respiratory Score (RS)** of 0–12 using four age-stratified variables (respiratory rate, retractions, dyspnea, auscultation), each scored 0–3. (seattlechildrens) Texas Children's uses a 6-component **Clinical Respiratory Score (CRS)** incorporating respiratory rate, auscultation, accessory muscles, mental status, SpO2, and color. (texaschildrens) (texaschildrens) Nationwide Children's employs a locally developed **Asthma Clinical Score (ACS)**, which a 2024 validation study showed comparable to PRAM (AUROC 0.77 vs. 0.74 for predicting disposition). (PubMed Central) Children's Mercy Kansas City uses a 5-element **Pediatric Asthma Score (PAS)** (Children's Mercy Kansas City) with validated thresholds: mild ≤7, moderate 8–11, severe ≥12. (Johns Hopkins Medicine) The **Pediatric Respiratory Assessment Measure (PRAM)** has the most robust external validation (Ducharme 2008) and is widely cited, (PubMed Central) though fewer U.S. institutions use it as their primary tool. (Medscape Reference)

The dexamethasone shift is now near-universal. Cincinnati Children's BESt 213 (2018) provided strong evidence that two doses of dexamethasone 0.6 mg/kg (one in the ED, one at home the next day) are equivalent to 5-day prednisone/prednisolone courses for relapse rates, return visits, and hospitalizations — with superior tolerance, compliance, and cost. (cincinnatichildrens) CHOP, Seattle Children's, Texas Children's, and most other institutions have adopted this approach. Seattle caps dexamethasone at 16 mg; most pathways use 0.6 mg/kg. The CHOP pathway (revised March 2025) includes an RN standing order for dexamethasone at triage for all severity levels. (Children's Hospital of Philadelp…)

Bronchodilator protocols follow a score-driven escalation. For mild disease, all pathways recommend albuterol MDI with spacer (4–8 puffs). For moderate disease, weight-based MDI dosing or nebulized albuterol every 20 minutes for three doses with the addition of ipratropium (up to 3 doses). For severe disease, continuous albuterol nebulization — Seattle uses 20 mg/hr via large-volume nebulizer or 7.5 mg via vibrating mesh; seattlechildrens CHOP doses by weight (5–10 kg = 7.5 mg/hr, >10–20 kg = 11.25 mg/hr, >20 kg = 15 mg/hr); Children's Hospital of Philadelp… Nationwide uses weight-stratified 1-hour bursts (15 mg if >15 kg, 10 mg if <15 kg). Nationwide Children's Hospital **IV magnesium sulfate at 40–50 mg/kg (max 2,000 mg)** is standard for severe non-responders Children's Hospital of Philadelp… in the first hour, with Seattle and CHOP using 50 mg/kg and Texas Children's using 40 mg/kg. Seattle Children's Seattle Children's

Disposition criteria are consistently score-driven. Seattle discharges at RS 1–4 sustained for ≥1 hour (2 hours if initial RS was 10–12), with oral tolerance and follow-up established. seattlechildrens Children's Mercy defines PICU criteria as continuous albuterol >4 hours with worsening, PCO2 >45, HFNC/NIV requirement, or SpO2 <90% on >2L/>50% FiO2. Children's Mercy Kansas City CHOP escalates to PICU when PASS ≥4 with FiO2 >50% or PASS = 6 with rising CO2. Children's Hospital of Philadelp…

The landmark **Kaiser et al. (2020) multisite study** across 83 EDs (37 children's hospitals, 46 community hospitals; n = 22,963) demonstrated that asthma pathway implementation significantly increased corticosteroid administration within 60 minutes (aOR 1.26), increased severity assessment at triage (aOR 1.88), and decreased admission/transfer rates — across both academic and community settings. PubMed The AAP's **PIPA (Pathways to Improve Pediatric Asthma) ED Toolkit** provides implementation materials specifically designed for this scale-up.

Freely available asthma pathway documents: CHOP ED pathway (chop.edu/clinical-pathway/asthma-emergent-care-clinical-pathway), Children's Hospital of Philadelp… Seattle Children's v12.2 (seattlechildrens.org/globalassets/documents/healthcare-professionals/clinical-standard-work/asthma_pathway.pdf), Texas Children's (texaschildrens.org), Nationwide Children's (nationwidechildrens.org), Johns Hopkins All Children's, Children's Mercy, and UNC Children's.

Bronchiolitis: the definitive case for de-implementation pathways

Bronchiolitis pathways represent perhaps the strongest example of evidence-based de-implementation in pediatric emergency medicine — systematically reducing interventions proven to be ineffective. The **AAP 2014 Clinical Practice Guideline** (Ralston et al., Pediatrics 2014;134(5):e1474–e1502, freely available at publications.aap.org) remains the current national standard. American Academy of Pediatrics PubMed No formal 2023 revision has been published; the AAP's 2023–2025 updates address RSV prevention (nirsevimab, palivizumab prophylaxis) rather than acute management.

The 2014 guideline's key action statements form the backbone of every institutional pathway: PLOS diagnose bronchiolitis clinically without routine labs or imaging; PubMed Central **do not administer albuterol, epinephrine, systemic corticosteroids, or chest physiotherapy**; texaschildrens do not use nebulized hypertonic saline in the ED (may consider for hospitalized patients); Stanford Medicine Children's H… administer supplemental oxygen only if SpO2 falls below 90%; and use nasogastric or intravenous fluids when oral hydration fails. Guideline Central These "do not" recommendations carry the guideline's strongest evidence ratings.

Institutional pathways uniformly reinforce these de-implementation targets while adding practical severity assessment and disposition frameworks. **CHOP's bronchiolitis pathway** (revised August 2025) classifies severity as mild/moderate/severe using descriptive criteria, initiates high-flow nasal cannula (HFNC) at 1.5 L/kg/min for severe patients not improving after suctioning, and escalates to CPAP at 8 cmH2O/0.4 FiO2 with PICU consultation if FiO2 exceeds 0.4. chop **Texas Children's** (October 2021) uses their Clinical Respiratory Score with HFNC initiation for CRS ≥7 or inability to maintain SpO2 ≥90% on 2 LPM. texaschildrens **Seattle Children's** (v17.0, September 2025) made a significant change in its latest version, **lowering the supplemental oxygen threshold from 92% to 90%**, Seattle Children's with SpO2 drops to 88% acceptable during sleep and brief drops to the 80s (<20 seconds) in sleeping children not requiring intervention. Seattle Children's Nwhrn

HFNC protocols have evolved substantially. Four randomized controlled trials demonstrated no clinical benefit of early routine HFNC versus low-flow nasal cannula in mild-to-moderate bronchiolitis — no differences in intubation rates or ICU utilization, with one study reporting longer stays with HFNC. American Academy of Pediatrics Evidence now supports HFNC as **rescue therapy after low-flow failure rather than routine use**. PubMed Central Lurie Children's (March 2023) codifies this as a "High Flow Pause" — mandating supportive care (nasal suction, soothe/swaddle, antipyretics, hydration optimization) before considering HFNC. luriechildrens The AAP VIP Network's **HI-FLO collaborative (Byrd et al., Pediatrics 2024)** demonstrated that a multicenter QI initiative using "Pause" (reduce HFNC initiation) and "Holiday" (expedite HFNC weaning) strategies reduced HFNC initiation and duration by over 30% without increasing length of stay. American Academy of Pediatrics ResearchGate

Nationwide Children's "Rest is Best" campaign, now in its fifth bronchiolitis season, exemplifies the de-implementation approach at scale: Children's Hospital Colorado it aligns care with AAP 2014 recommendations across ED, urgent care, and inpatient settings, Nationwide Children's Hospital has achieved significant reductions in unnecessary interventions with no increase in ICU admissions or 7-day readmissions, and actively discourages routine viral testing — noting that parents do not expect it and that labeling with "RSV" rather than "bronchiolitis" increases family anxiety without changing management. Nationwide Children's Hospital

Quality improvement outcomes from pathway implementation are robust. Ralston et al.'s 2014 systematic review of 14 QI studies (>12,000 infants) showed pathways reduced bronchodilator use by 16/100 patients, steroids by 5/100, chest X-rays by 9/100, and antibiotics by 4/100. PubMed Vanderbilt's high-reliability interventions (bronchiolitis-specific order sets, EHR best practice alerts) achieved sustained CXR reduction from 42.1% to 18.9% without increasing 72-hour returns. American Academy of Pediatrics PubMed The VIP B-QIP collaborative across 17 centers achieved a **46% reduction in bronchodilator use** among 11,568 hospitalizations. Journalofhospitalmedicine

Additional freely available bronchiolitis pathways: Lurie Children's ED algorithm (luriechildrens.org), Johns Hopkins All Children's (hopkinsmedicine.org), Johns Hopkins Medicine Children's Mercy Kansas City synopsis (July 2023), Childrensmercy Children's Minnesota, OHSU/Doernbecher, and UNC Children's.

Febrile infant evaluation: the AAP 2021 guideline transforms a generation of practice

The **AAP 2021 Clinical Practice Guideline for well-appearing febrile infants 8–60 days old** (Pantell et al., Pediatrics 2021;148(2):e2021052228, freely available at publications.aap.org) represents the most significant change in febrile infant management in decades, replacing historical criteria (Rochester, Philadelphia, Boston) with age-stratified algorithms that integrate **procalcitonin** as a central biomarker and incorporate shared decision-making for the first time.

The guideline applies to well-appearing, term (≥37 weeks) infants with rectal temperature ≥38.0°C within 24 hours, Cincinnati Children's Hospital and provides three age-stratified algorithms PubMed with 21 key action statements. **For infants 8–21 days old**, all require urinalysis, urine culture, blood culture, inflammatory markers (procalcitonin >0.5 ng/mL, CRP >20 mg/L, ANC >4,000–5,200/mm³), CSF analysis and culture, parenteral antimicrobials, and hospitalization — regardless of laboratory results. **For infants 22–28 days old**, the guideline recommends full laboratory evaluation including inflammatory markers; CSF should be obtained if any inflammatory marker is abnormal, but if all are normal, the landmark recommendation is **shared decision-making with parents regarding lumbar puncture** — a major departure from prior universal LP practice. PubMed Home management becomes possible with intramuscular ceftriaxone, written safety-net instructions, and 24-hour follow-up if CSF is normal and UA is negative. **For infants 29–60 days old**, CSF is recommended only if more than one inflammatory marker is abnormal, and oral antibiotics are acceptable for urinalysis-positive infants with normal inflammatory markers — both significant practice changes.

PECARN febrile infant research has both validated and extended these guidelines. The PECARN Febrile Infant Rule (Kuppermann et al., JAMA Pediatrics 2019) derived from 1,896 febrile infants ≤60 days across 26 EDs Academic Life in EM identified a three-variable low-risk rule — negative urinalysis, ANC ≤4,090/mm³, and procalcitonin ≤1.71 ng/mL — with **sensitivity of 97.7% and NPV of 99.6%** for serious bacterial infection. WikEM A 2025/2026 PECARN study of febrile infants ≤28 days demonstrated that the combination of negative UA, ANC ≤4,000, and PCT ≤0.5 identified a low-risk group with zero bacterial meningitis, supporting shared decision-making for LP in over 40% of febrile infants in the first month. Emergency Medicine Cases For infants **61–90 days old** — a gap not covered by the AAP guideline — Aronson et al. (Pediatrics 2025) derived two prediction rules from 4,952 infants: American Academy of Pediatrics Rule 1 (negative UA + Tmax ≤38.9°C; sensitivity 86%, NPV 99.5%) and Rule 2 (PCT ≤0.24 + ANC ≤10,710; sensitivity 100%, specificity 66%), though these require prospective validation.

Institutional pathways have rapidly incorporated the AAP 2021 framework with institution-specific modifications. **CHOP** (chop.edu/clinical-pathway/febrile-infant-emergent-evaluation-clinical-pathway) covers infants ≤56 days Children's Hospital of Philadelp… chop with integrated EHR clinical decision support and published QI results showing LP rates in 22–28 day olds decreased from 87% to 44% — sustained over 2 years with zero missed bacterial meningitis cases. **Nationwide Children's** publishes separate ED pathway PDFs for each age stratum Nationwide Children's Hospital (0–21, 22–28, and 29–60 days), revised April 2024, with the notable guidance that "mild, isolated pleocytosis is NOT an indication for routine acyclovir." Nationwide Children's Hospital **Texas Children's** (August 2023) takes a more restrictive approach to HSV, recommending empiric acyclovir only with cutaneous lesions. **Seattle Children's** (v11.0, October 2024) uses GRADE-rated evidence with procalcitonin prechecked in order sets for 22–60 day olds. **UCSF Benioff** extends coverage to 0–90 days, addressing the AAP guideline gap and noting that virus-positive infants <29 days still carry a 3–13% invasive bacterial infection rate.

The **REVISE II national QI collaborative** (McDaniel et al., Pediatrics 2024) deployed the AAP 2021 guideline across 103 hospitals with multifaceted interventions including education, pathway updates, clinical decision support, and order sets — achieving significant improvements in guideline-concordant care. American Academy of Pediatrics American Academy of Pediatrics However, an equity analysis of 16,961 infants across 99 U.S. hospitals identified persistent racial/ethnic disparities in guideline-concordant care even after QI intervention. American Academy of Pediatrics

For community EDs, procalcitonin availability remains the critical implementation barrier. The AAP guideline and PECARN rules depend on procalcitonin for optimal risk stratification, yet many community hospitals lack point-of-care or rapid-turnaround PCT assays. Workaround strategies include using CRP >20 mg/L and ANC >4,000 as surrogates (less specific) or advocating for send-out PCT testing with results available within hours. The Alaska Native Medical Center (anmc.org/files/fev1.pdf) and Yukon-Kuskokwim Health Corporation pathways provide models specifically designed for remote settings with limited laboratory capabilities. Yk-health

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Pediatric trauma: PECARN rules now prospectively validated for community implementation

The cornerstone of evidence-based pediatric trauma pathway design is the set of **PECARN clinical decision rules**, which were prospectively validated in a 2024 study of 19,999 children at 6 Level-1 pediatric trauma centers (Lancet Child & Adolescent Health 2024;8(5):339–347) PubMed — confirming their outstanding test characteristics and suitability for broad implementation. PubMed ScienceDirect

The PECARN Head Injury/TBI Prediction Rule (Kuppermann et al., Lancet 2009; n = 42,412) provides separate age-stratified algorithms. Medscape Reference Pecarn For **children under 2 years**, high-risk criteria mandating CT include GCS <15, palpable skull fracture, or altered mental status — carrying a 4.4% ciTBI risk. Medscape Reference Medicqi Intermediate-risk criteria (0.9% risk) include non-frontal scalp hematoma, loss of consciousness >5 seconds, not acting normally per parent, or severe mechanism (fall >3 feet, MVC with ejection/rollover/fatality, struck by high-impact object); these warrant observation versus CT with shared decision-making. If no criteria are present, ciTBI risk is **<0.02%** and CT is not indicated. medicqi Medscape Reference For **children ≥2 years**, high-risk criteria are GCS <15, signs of basilar skull fracture, or altered mental status (4.3% risk). Intermediate-risk criteria (0.8% risk) include vomiting, loss of consciousness, severe headache, or severe mechanism (fall >5 feet). medicqi The 2024 validation confirmed sensitivity of **100% for children <2 years** and **98.8% for children ≥2 years**, Medscape Reference with two misclassified older children requiring only observation without neurosurgery. ScienceDirect PubMed

The PECARN Intra-Abdominal Injury Rule (Holmes et al., Ann Emerg Med 2013) uses seven clinical variables — all obtainable from history and physical examination without labs or imaging: Thesgem ScienceDirect abdominal wall trauma/seatbelt sign, GCS <14, abdominal tenderness, thoracic wall trauma, abdominal pain, decreased breath sounds, and vomiting. PubMed Central If all seven are negative, CT can safely be avoided. PubMed Central The 2024 validation confirmed **sensitivity of 100% and NPV of 100%** for injuries requiring intervention. ScienceDirect PubMed A 2025 "grey zone" analysis (Arnold et al., Academic Emergency Medicine) characterized risk among children with 1–2 positive variables: IAI-intervention rates were 1.3% with one variable and 2.0% with two, with GCS <14 carrying the highest individual risk at 5.5%. PubMed The rule should not be applied to suspected non-accidental trauma cases. PubMed

CHOP publishes the most comprehensive publicly available **pediatric head trauma pathway** (chop.edu/clinical-pathway/head-trauma-acute-clinical-pathway, revised May 2025), integrating PECARN criteria directly into clinical workflow with a visio-vestibular exam for concussion screening in children >5 years and automatic social work consult with physical abuse pathway activation for all children <2 years. chop Disposition criteria specify PICU admission for epidural/subdural hemorrhages ≥5 mm with neurologic symptoms. chop The pathway also links to a modified pediatric Glasgow Coma Scale reference with infant-specific verbal and motor subscale modifications. chop Children's Hospital of Philadelp…

Nationwide Children's publishes the most detailed publicly available **non-accidental trauma (NAT) pathway** (revised 2023), with age-based screening triggers and workup recommendations: skeletal survey plus head imaging plus laboratory screening (LFTs, lipase, UA) for children <2 years, SAEM with the **TEN-4-FACESp bruising rule** as a screening tool — any bruising on the torso, ears, or neck in children <4 years, or any bruise in infants ≤4 months, triggers evaluation. EMSC Improvement Center

For concussion management, the **CDC Pediatric mTBI Guideline (2018)** provides 19 recommendation sets CDC PubMed Central emphasizing validated clinical decision rules over routine imaging, CDC brief cognitive rest (1–2 days maximum), gradual return to activity, CDC American Academy of Family … and structured **return-to-learn before return-to-play** protocols. The AAP/International Consensus 6-step return-to-play progression requires ≥24 hours at each step: PubMed Central (1) symptom-limited activity, (2) light aerobic exercise, (3) sport-specific exercise without head impact, (4) non-contact training drills, (5) full-contact practice after medical clearance, (6) return to competition.

Community hospital transfer criteria for pediatric trauma are well-defined: immediate transfer for GCS ≤13 or deteriorating neurologic status, any intracranial hemorrhage on CT, depressed/compound skull fracture, spinal cord injury, pelvic fracture, ≥2 long bone fractures, hemodynamic instability not responsive to initial resuscitation, or suspected child abuse requiring specialist evaluation. Transfer timeline goals are within 1 hour for emergencies and 4 hours for urgent cases, CA using pediatric specialty transport when available Emscimprovement and forwarding imaging electronically. Emscimprovement

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National infrastructure powering pathway dissemination and quality improvement

Several national organizations form the backbone of pediatric ED pathway dissemination. **PECARN** (pecarn.org), federally funded through HRSA's EMSC Program, has produced the field's most validated clinical decision rules, all freely available on MDCalc. The **AAP Value in Inpatient Pediatrics (VIP) Network** AAP has executed major multi-center QI initiatives American Academy of Pediatrics American Academy of Pediatrics — B-QIP for bronchiolitis (17 centers, 46% bronchodilator reduction), Journalofhospitalmedicine PIPA for asthma (68 institutions), American Academy of Pediatrics REVISE for febrile infants (103 hospitals), American Academy of Pediatri… HI-FLO for HFNC reduction, ResearchGate and BASiC for antibiotic stewardship (118 hospitals) American Academy of Pediatrics American Academy of Pediatrics — with the critical strength of including non-academic and adult-serving hospitals. AAP

The **National Pediatric Readiness Project** (pedsready.org) provides the most important infrastructure for community EDs, with a free 7-domain toolkit, self-assessment with gap analysis, and QI dashboards. A 2023 systematic review of 8 studies confirmed that higher pediatric readiness scores correlate with up to **76% lower mortality risk** among critically ill and injured children, at a cost of just $4–$48 per pediatric patient. A new 2026 joint policy statement from AAP, ACEP, ENA, and ACS updates readiness standards with increased focus on clinical practice guidelines and decision support tools. ENA

The **2022 Choosing Wisely Pediatric Emergency Medicine recommendations** (Mullan et al., Ann Emerg Med 2024) provide five specific targets for reducing low-value care: PubMed ScienceDirect do not obtain radiographs in bronchiolitis, croup, asthma, or first-time wheezing; Choosing Wisely Canada do not obtain screening labs for psychiatric medical clearance; do not obtain labs or head CT for unprovoked generalized or simple febrile seizures with return to baseline; do not obtain abdominal radiographs for constipation; and do not routinely test for respiratory viruses when results will not change management. ScienceDirect A 2025 adherence study of 6.7 million visits found lower-volume EDs are more likely to perform unnecessary testing ScienceDirect — precisely the settings this guide targets.

The **Children's Hospital Association** maintains the Pediatric Health Information System (PHIS), a comparative database covering inpatient, ambulatory, ED, and observation encounters from 50+ leading children's hospitals with 2,500+ pediatric-specific data points, Children's Hospital Association plus benchmarking tools including PREP (readmission predictor for 324 diagnosis groups) Children's Hospital Association and a Low-Value Care Calculator with 16 measures. American Academy of Pediatrics

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Practical implementation roadmap for community and rural EDs

Community and rural EDs face specific barriers — limited pediatric subspecialty access, restricted laboratory capabilities (especially procalcitonin), lower case volumes reducing pattern recognition, and equipment/medication gaps. Several resources directly address these challenges.

Telemedicine partnerships represent the highest-impact intervention. Primary Children's Hospital (Intermountain) provides 24/7 telemedicine consultation plus training, simulations, and evidence-based protocol binders to rural hospitals across five states, with partner hospitals raising readiness scores by 25+ points within 2 years. Children's Hospital Association UC Davis's pediatric critical care telemedicine program connecting 8 rural Northern California EDs demonstrated that telemedicine consultations reduced hospital admissions compared to telephone consultations (59.5% vs. 87.5%). PubMed Central

Cincinnati Children's Pediatric Trauma Training Center (PTTC) offers a partnership model for community hospitals that includes online curriculum (10 hours CME/year) and substitutes the requirement for an on-site pediatric surgeon with standardized protocols and transfer agreements — providing an alternative pathway to ACS trauma verification. Cincinnati Children's

LA Pediatric Readiness (lapedsready.org/index.php/clinical-pathways/) aggregates and links to pathways from CHOP, Seattle Children's, and other institutions specifically for community hospital use. The **Alaska Native Medical Center** and **Yukon-Kuskokwim Health Corporation** publish febrile infant pathways designed for remote settings Alaska Native Medical Center — models directly applicable to rural continental U.S. hospitals.

For digital clinical decision support, **MDCalc** (free; mdcalc.com) hosts all major PECARN rules, Rochester criteria, and pediatric scoring tools — used weekly by 65% of U.S. attending physicians. Wikipedia EB Medicine **PEMCalc** (free; pemcalc.com) provides pediatric-specific calculators including WETFLAG resuscitation doses, weight-based drug dosing, age-specific vital sign ranges, and BP percentiles optimized for mobile bedside use. Pemcalc

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Conclusion

The evidence base for pediatric ED care pathways has reached a level of maturity and accessibility that makes standardized, guideline-concordant care achievable in virtually any emergency department setting. The most critical insight for community hospitals is that **the pathway infrastructure already exists and is freely available** — what remains is local adaptation, implementation, and the establishment of pediatric telemedicine partnerships.

Three strategic priorities emerge for rural and community ED protocol development. First, **adopt PECARN clinical decision rules as the foundation** for trauma and febrile infant evaluation — these are the most validated tools in pediatric emergency medicine, designed for broad applicability, and require no specialized equipment beyond standard laboratory testing. Second, **adapt rather than create de novo** by selecting one comprehensive pathway source (CHOP for breadth, Seattle for outcomes evidence, Texas Children's for GRADE rigor) and systematically localizing their protocols to available resources. Third, **engage with the National Pediatric Readiness Project** and regional children's hospital outreach programs, which provide the scaffolding — equipment checklists, competency standards, QI frameworks, and teleconsultation networks — that transforms a pathway document into sustainable practice change.

The persistent gap is not in evidence or pathway availability but in implementation support, laboratory access (particularly procalcitonin for febrile infant evaluation), and systematic measurement of outcomes. The VIP Network's success in deploying QI collaboratives across 100+ hospitals — including non-academic and adult-serving facilities American Academy of Pediatrics — demonstrates that these barriers are surmountable PubMed when pathway implementation is supported by structured collaborative infrastructure American Academy of Pediatrics American Academy of Pediatrics rather than left to individual institutional initiative. UCSF